

Unit v

Ethics of AI in Transport

Introduction

- Artificial Intelligence is changing transportation through:
 - self-driving cars
 - smart traffic systems
 - AI-based public transport
 - delivery drones
 - accident prediction systems
- Although AI improves safety and efficiency, it also raises important ethical questions.

- **1. Safety and Human Life**
 - The first ethical issue is **safety**.
 - AI transport systems make decisions in real time.
For example:
 - A self-driving car may need to decide between:
 - protecting passengers
 - protecting pedestrians
 - avoiding a collision
- This creates a moral question:
- **Who should the AI protect first?**
 - If an accident happens:
 - Is the car manufacturer responsible?
 - Is the software developer responsible?
 - Or is the passenger responsible?
 - Safety must always be the highest priority.

- **2. Accountability**

- The second issue is **accountability**.

- When humans drive, responsibility is clear.
But with AI systems, responsibility becomes complicated.

- For example:

If an autonomous bus crashes because of a software error, who is legally responsible?

- Ethical transport systems need:

- clear laws
 - transparent decisions
 - human oversight

- **Privacy Concerns**
- AI in transport collects large amounts of personal data such as:
 - location
 - travel habits
 - facial recognition
 - voice commands
- This raises privacy concerns because people may not know:
 - what data is collected
 - how it is stored
 - who can access it
- Ethically, companies must:
 - protect user data
 - ask consent
 - prevent misuse

- **Bias and Fairness**

- AI systems can sometimes show **bias**.

- For example:

An AI traffic system may prioritize wealthy neighborhoods while ignoring poorer areas.

- This can create:

- unfair service
- unequal access
- social discrimination

- Ethical AI should treat all communities fairly.

- **Employment Impact**
 - AI may replace workers such as:
 - taxi drivers
 - truck drivers
 - delivery workers
- This creates ethical concerns about:
 - job loss
 - economic inequality
 - social disruption
- Society must support workers through:
 - retraining
 - education
 - new job opportunities

- To conclude,
- AI in transport can make travel:
 - safer
 - faster
 - smarter
- But it must also be:
 - fair
 - safe
 - transparent
 - human-centered
- Ethics is essential to ensure technology benefits everyone.

Ethics of AI in Biomedical Research

- AI is helping scientists in:
 - disease diagnosis
 - drug discovery
 - gene analysis
 - medical imaging
 - personalized medicine
- However, using AI in healthcare research creates serious ethical challenges.

- **Patient Privacy**
- The first ethical issue is **patient privacy**.
- Biomedical AI uses sensitive data such as:
 - medical records
 - DNA information
 - scan reports
 - health history
- If this data is leaked, patients may face:
 - discrimination
 - stigma
 - identity theft
- Researchers must ensure:
 - strong data protection
 - informed consent
 - confidentiality

- **Informed Consent**
- Patients should understand:
 - how their data is used
 - why AI is analyzing it
 - what risks are involved
- Sometimes AI systems are complex, making it difficult for patients to understand.
- Ethically, researchers must explain AI in simple language before collecting data.

- **Bias in Medical AI**

- AI systems learn from data.
- If data contains bias, AI may produce unfair results.
- For example:
 - AI may diagnose some diseases better in one group than another because of unequal training data.
- This can lead to:
 - misdiagnosis
 - unequal treatment
 - healthcare discrimination
- Ethical biomedical AI must use diverse and balanced data.

- **Transparency**
- Many AI systems work like a “black box.”
- This means doctors may not know:
 - how the AI reached a decision
 - why a patient was classified as high-risk
- This creates trust problems.
- Ethical AI should be:
 - explainable
 - understandable
 - transparent
- Doctors should be able to explain AI decisions to patients.

- **Human Control**
- AI can assist doctors, but should not fully replace them.
- Medical decisions involve:
 - compassion
 - judgment
 - empathy
- Machines cannot completely replace human care.
- Ethically:
 - AI should support doctors, not replace human responsibility.

- **Ownership of Medical Data**
- Another issue is data ownership.
- Questions include:
 - Who owns patient data?
 - The hospital?
 - The researcher?
 - The company?
 - Or the patient?
- Most ethicists believe:
Patients should have control over their own medical data.

- AI can improve biomedical research by:
 - finding diseases earlier
 - developing medicines faster
 - improving patient care
- But it must respect:
 - privacy
 - fairness
 - transparency
 - human dignity
- Ethics ensures that AI serves humanity responsibly.